

Blood Donor Management System - Full Project Plan

1. Tech Stack

- Backend: Golang (Gin/Fiber)
- ORM: GORM
- Database: PostgreSQL
- Frontend: Vue.js 3
- Authentication: JWT (Access + Refresh)
- Cache: Redis
- Deployment: Docker + Nginx

2. Role Based System

- Admin: Full system control, manage users, view logs, analytics
- Manager: Manage donors, regional control
- Donor: Create/update profile, manage availability, view phone logs

3. Core Features

- Public donor search by blood group and location
- Login required to view phone number
- Phone view logging system
- Search logging system
- Donation cooldown logic (3-4 months rule)
- Privacy control for donors

4. Security Features

- JWT authentication
- Role-based middleware
- Password hashing (bcrypt)
- Rate limiting
- Activity audit logs
- Input validation & SQL injection prevention via GORM

5. Database Tables (GORM Models)

- Users (id, name, email, phone, password, role, status, created_at)
- Donors (id, user_id, blood_group, last_donation_date, district, thana, lat, long, available)
- PhoneViewLogs (id, donor_id, viewed_by_user_id, viewed_at, ip_address)
- SearchLogs (id, search_by_user_id, blood_group, location, searched_at)

6. Advanced Features (Phase 2+)

- Emergency blood request system
- Geo-location based nearest donor search (Haversine formula)
- SMS notification integration
- Analytics dashboard
- Blacklist & IP blocking
- Hospital blood stock management

7. Development Phases

- Phase 1: Auth + Donor CRUD + Search + Logging
- Phase 2: Emergency system + Geo search + Analytics
- Phase 3: SMS + Hospital panel + Mobile app

8. Scalability Plan

- Use Redis for caching search results
- Background workers for notifications
- Docker-based deployment
- Future microservice architecture